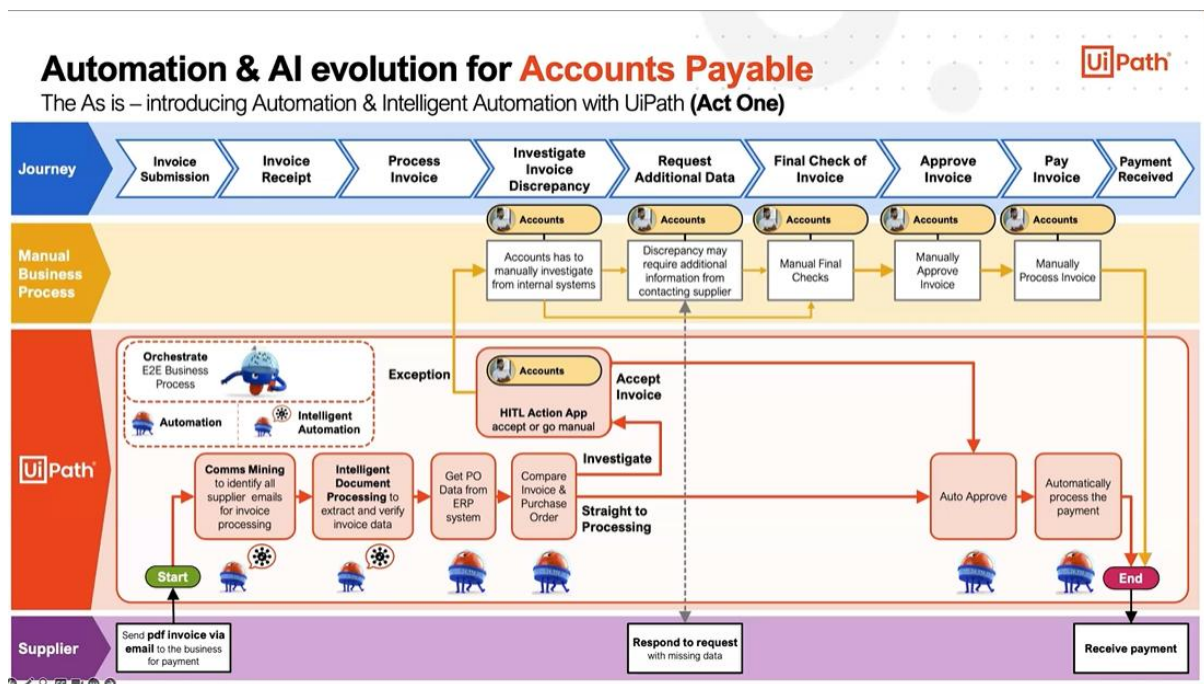


AI Accounts Payable (AP) Employee

Technical Design Document:

Topic covers

- ✓ Topic 1: Problem Framing & Goals
- ✓ Topic 2: Assumptions & Constraints
- ✓ Topic 3: System Overview
- ✓ Topic 4: Multi-Agent Architecture
- ✓ Topic 5: Agent Loop & Orchestration
- ✓ Topic 6: Core Workflows
- ✓ Topic 7: Tooling & Integration Layer
- ✓ Topic 8: Memory & Context Management
- ✓ Topic 9: Observability & Auditability
- ✓ Topic 10: Data Models (High-Level)
- ✓ Topic 11: Failure Modes & Recovery
- ✓ Topic 12: MVP Scope & What Is Deferred



➤ Topic 1: Problem Framing & Goals

What Accounts Payable Is

Accounts Payable (AP) is the finance function responsible for receiving supplier invoices, matching them to purchase orders and receipts when applicable, ensuring policy compliance (VAT, vendor validation, duplicate checks, correct GL coding), routing invoices for approval, managing payment scheduling and vendor communication, maintaining segregation of duties and regulatory compliance, and optimizing cash flow.

Why AP Is Complex and Full of Edge Cases

AP workflows involve numerous complex scenarios that require contextual understanding:

- **Missing Purchase Orders:** Invoices arrive without corresponding POs, requiring procurement verification
- **Partial Receipts:** Goods received in multiple shipments, requiring careful reconciliation
- **Price/Quantity Mismatches:** Variations between PO, receipt, and invoice requiring tolerance validation
- **Duplicate Invoices:** Identical invoices from same vendor requiring detection and prevention
- **Supplier Changes:** Bank details updates presenting fraud risk
- **Multi-Currency Transactions:** Exchange rate fluctuations affecting amounts
- **VAT Issues:** Tax validation and compliance across jurisdictions
- **Urgent Exceptions:** Critical suppliers requiring expedited processing
- **Fraud Risk:** Suspicious patterns requiring investigation

Why Autonomous Agents (Not Chatbots) Are Required

Traditional chatbot approaches provide "middle-to-middle" assistance, requiring constant human oversight. True task completion demands autonomous agents that can:

- Process invoices end-to-end without human intervention
- Make contextual decisions based on business rules
- Handle exceptions and edge cases independently

Goals of the AI AP Employee

- **End-to-End Automation:** Process 90%+ of invoices without manual intervention
- **Sub-2-Hour Processing:** From receipt to payment approval
- **Exception Handling:** Autonomous resolution of 80%+ of common exceptions
- **Compliance Assurance:** 100% adherence to regulatory requirements
- **Cross-Company Generalization:** Plug-and-play deployment across organizations

➤ Topic 2: Assumptions & Constraints

Company-Agnostic, Safe, and Regulation-Aware System Design

Company-Agnostic Architecture

- Designed to operate across multiple organizations without custom code changes.
 - Uses configuration-driven workflows to adapt to different accounting policies and approval rules.
 - Supports diverse organizational structures through flexible, role-based approval hierarchies.
 - Enables seamless deployment across companies of varying sizes and industries.
 - Minimizes onboarding friction through standardized configuration templates.
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No Hardcoded Workflows

- Business logic is stored in a centralized, configurable rule engine.
 - Workflow behavior changes dynamically based on company-specific policies.
 - Approval routing is determined by invoice value, vendor risk, and tax impact.
 - Abstracted tool interfaces allow easy replacement or addition of integrations.
 - Plug-and-play connectors support different ERP and accounting platforms.
 - Policy inheritance allows global rules with localized overrides for subsidiaries.
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Safety-First Autonomous Execution

- Implements human-in-the-loop approval gates for critical financial decisions.
 - Restricts autonomous execution for high-risk actions such as bank detail changes.
 - Maintains complete, immutable audit logs for every agent action.
 - Records decision context, triggered rules, and responsible agents.
 - Provides rollback mechanisms to safely undo incorrect operations.
 - Gradually increases autonomy based on accuracy and compliance metrics.
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Regulatory Awareness and Compliance

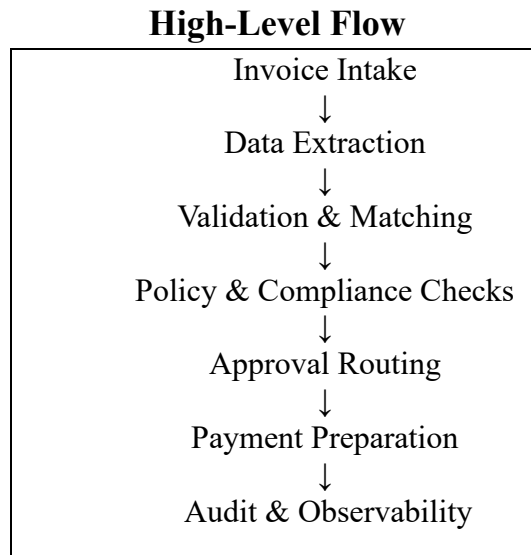
- Enforces VAT validation aligned with UK and EU regulatory requirements.
 - Verifies invoice tax rates, vendor VAT IDs, and document completeness.
 - Implements SOX-compliant segregation of duties and approval controls.
 - Applies GDPR principles through data minimization and PII redaction.
 - Restricts sensitive data access using role-based permissions.
 - Supports extension for industry regulations such as PCI-DSS and regional tax laws.
-

Technical Constraints

- External systems such as ERP platforms or email services may have temporary failures.
- AI outputs may contain uncertainty; therefore, deterministic validation and human escalation are required.

➤ Topic 3: System Overview

The AI AP Employee is composed of multiple specialized agents orchestrated through a central execution loop. The system continuously ingests invoices, validates them, routes approvals, handles exceptions, and prepares payment instructions while logging every decision.



AI Accounts Payable Employee

- A multi-agent autonomous system designed to execute the Accounts Payable process end-to-end.
- Simulates the responsibilities of a trained AP professional with built-in safety and compliance controls.
- Operates continuously in the background with minimal human supervision.
- Processes invoices from intake to payment preparation while maintaining full auditability.
- Learns from historical decisions to improve accuracy and efficiency over time.

Major System Components

- Agent Orchestrator
 - Acts as the central coordinator for all agent activities.
 - Manages workflow state, task sequencing, and error recovery.
 - Enforces approval gates and policy constraints.
- Specialized Agents
 - Invoice Intake Agent for document ingestion and classification.
 - Data Extraction Agent for structured data parsing from invoices.
 - Validation & Matching Agent for duplicate checks and 2-way / 3-way matching.
 - Compliance Agent for VAT, policy, and regulatory validation.
 - Communication Agent for vendor and approver interactions.

- Memory System
 - Structured memory stores invoices, vendors, approvals, and audit logs.
 - Semantic memory captures patterns, past exceptions, and resolution strategies.
 - Tool Integration Layer
 - Abstracted connectors for ERP systems, email, messaging, and banking platforms.
 - Audit & Compliance Engine
 - Records every action with timestamps, reasons, and triggered rules.
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End-to-End Autonomous Execution

- Executes self-directed agent loops with defined success and failure conditions.
- Uses decision trees to handle exceptions such as missing POs or mismatches.
- Monitors SLAs, due dates, and payment schedules proactively.
- Escalates to humans only when risk thresholds are exceeded.
- Adapts workflows dynamically using configuration-driven company policies.

➤ Topic 4: Multi-Agent Architecture

Overview

- The AI AP Employee is composed of multiple specialized agents coordinated by a central orchestrator.
 - Each agent owns a clearly defined responsibility within the Accounts Payable workflow.
 - Agents collaborate through shared memory, standardized messages, and event-driven signals.
 - This modular design improves scalability, fault isolation, and generalizability across companies.
-

Invoice Capture (Intake) Agent

- **Responsibility:** Ingest invoices and supporting documents from multiple sources.
 - **Inputs:** Email inbox attachments, document uploads, API payloads.
 - **Outputs:** Raw invoice documents with metadata (source, timestamp, sender).
 - **Collaboration**
 - Works with the Tool Integration Layer for email and document handling.
 - Notifies the orchestrator when new invoices are available.
-

Extraction Agent

- **Responsibility:** Extract structured data from invoice documents.
- **Inputs:** Invoice PDFs, scanned images, and raw documents.
- **Outputs:** Vendor details, invoice number, dates, amounts, tax, and line items.
 - Confidence scores for each extracted field.
- **Collaboration:** Interfaces with OCR and NLP processors.
 - Consults semantic memory for layout and pattern recognition.

Validation & Matching Agent

- **Responsibility:** Validate invoice accuracy and consistency.
 - **Checks:** Duplicate invoice detection.
 - Two-way and three-way matching against purchase orders and goods receipts.
 - **Outputs:** Validated invoice status or exception indicators.
 - **Collaboration:** Queries vendor master data and ERP systems via structured memory.
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Policy & Compliance Agent

- **Responsibility:** Enforce company policies and regulatory requirements.
 - **Checks:** VAT and tax validation and Vendor verification and segregation of duties.
 - **Outputs:** Compliance approval or policy exception flags.
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Approval Orchestration Agent

- **Responsibility:** Route invoices through appropriate approval workflows.
 - **Inputs:** Invoice state, approval rules, organizational hierarchy.
 - **Outputs:** Approval requests, decisions, or escalation alerts.
 - **Collaboration:** Integrates with notification and user management systems.
-

Payment Preparation Agent

- **Responsibility:** Prepare accounting entries and payment instructions.
 - **Inputs:** Fully approved invoices and verified vendor bank details.
 - **Outputs:** Payment-ready records without executing transactions.
 - **Collaboration:** Interfaces with ERP and banking APIs.
-

Audit & Observability Agent

- **Responsibility:** Maintain full system transparency and traceability.
 - **Inputs:** All system events, decisions, and agent actions.
 - **Outputs:** Immutable audit logs, metrics, alerts, and evidence trails.
 - **Collaboration:** Passively observes all agents to ensure compliance and accountability.
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➤ Topic 5: Agent Loop & Orchestration

Autonomous Execution Loop

- The system operates using a continuous autonomous execution loop.
 - Each invoice progresses through clearly defined phases to ensure reliability and safety.
 - The loop follows: **Plan → Act → Observe → Recover → Complete**
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Plan Phase

- Analyze the current invoice state and processing history.
 - Determine the next required actions based on workflow rules.
 - Consult structured and semantic memory for historical context.
 - Identify necessary tools, agents, and approval requirements.
 - Define success and failure criteria for the planned actions.
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Act Phase

- Execute planned actions using specialized agents.
 - Invoke external tools such as **OCR services**, **ERP systems**, or compliance checkers.
 - Apply company-specific policies and approval rules.
 - Update the workflow state after each action.
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Observe Phase

- Monitor outputs and responses from executed actions.
 - Collect feedback from integrated systems and agents.
 - Measure confidence scores, validation results, and response statuses.
 - Detect anomalies, delays, or unexpected outcomes.
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Recover Phase

- Trigger retry mechanisms when recoverable failures occur.
 - Apply fallback strategies such as alternative extraction or matching methods.
 - Escalate to human reviewers for high-risk or unresolved issues.
 - Safely pause processing when critical errors are detected.
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Complete Phase

- Finalize the invoice processing state.
 - Update audit logs with full decision context.
 - Trigger downstream steps such as payment preparation or reporting.
 - Mark the workflow as completed or safely blocked.
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State Machine Model

- Uses a finite state machine with defined transitions:
 - **RECEIVED → EXTRACTED → VALIDATED → MATCHED → APPROVED → PAID → COMPLETED**

Success and Failure Criteria

- **Success:** All validations pass and approvals are obtained.
- **Failure:** Critical policy violations or system unavailability.
- **Exception:** Expected edge cases requiring alternate workflows.

➤ Topic 6: Core Workflows

Happy Path (Clean Invoice with PO)

- Invoice received via email and classified correctly by intake agent
- OCR extracts invoice data with high confidence scores
- Three-way matching confirms PO, invoice, and receipt alignment
- Auto-approval triggered based on configured amount thresholds
- Payment prepared, scheduled, and logged with full audit trail

Missing PO

- Invoice received without purchase order reference detected automatically
 - Validation agent flags missing PO and pauses processing
 - Procurement team notified to provide PO or exception approval
 - Invoice placed on temporary hold pending confirmation
 - Workflow resumes once PO linkage or approval completed
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Duplicate Invoice Detection

- Incoming invoice compared against recent processed invoice records
 - Vendor, amount, date, and reference matched for duplication
 - Exact duplicate detected and flagged for human verification
 - Human reviewer confirms duplicate or valid resubmission
 - Invoice rejected or reinstated based on review outcome
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Three-Way Match Mismatch

- Discrepancy detected between invoice, PO, and receipt values
 - Exception workflow initiated with detailed variance explanation
 - Stakeholders notified to review mismatch and supporting documents
 - Manual approval requested for tolerance override decision
 - Adjustments recorded and invoice processing resumed
-

Vendor Bank Detail Change (Fraud Risk)

- Invoice includes request for updated vendor bank details
 - New details compared against existing vendor master records
 - High-risk change flagged and blocked automatically
 - Multi-factor verification initiated for vendor confirmation
 - Bank details updated only after successful verification
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VAT / Tax Validation Failure

- VAT calculations validated against jurisdiction-specific tax rules
 - Incorrect or missing VAT detected during compliance checks
 - Supplier notified to correct invoice tax details
 - Invoice placed on hold pending compliant resubmission
 - Processing resumes after successful VAT validation
-

Approval SLA Breach Risk

- SLA monitoring detects approval deadline approaching soon
 - Escalation workflow triggered based on policy rules
 - Alternate approvers notified for expedited review
 - Priority routing applied to prevent payment delays
 - Stakeholders informed of revised approval timeline
-

➤ Topic 7: Tooling & Integration Layer

Email Ingestion (Gmail/Outlook)

- Secure email access using IMAP or POP3 protocols.
- Automated PDF and image attachment processing for invoices.
- Security filters detect malware and phishing threats automatically.
- Metadata extracted for sender reputation and message history.
- New emails trigger workflow notifications to orchestrator agents.

Document Parsing (PDFs)

- OCR engine extracts text accurately from all invoice types.
- Layout analysis identifies structural elements and line-item tables.
- Fields intelligently mapped to structured invoice data.
- Confidence scores generated for each extracted field.
- Validation ensures accuracy before downstream processing.

ERP & Accounting Systems

- API abstraction enables connectivity with multiple ERP platforms.
- Flexible field mapping supports different accounting system schemas.
- Synchronization supports both real-time and batch updates.
- Retry mechanisms handle temporary failures robustly.
- Validation ensures consistency between ERP and agent memory

Messaging Tools (Slack/Teams)

- Alerts configured for approvals, escalations, and exceptions.
- Direct workflow integration allows interactive task handling.
- Users approve or escalate invoices via messaging interface.
- Real-time status updates communicated to stakeholders.
- Notifications trigger downstream agent actions automatically.

Banking/Payment Preparation

- Generates standardized payment files without executing transfers.
- Secure banking API integration ensures safe communication.
- Pre-payment validation checks verify account and invoice data.
- Scheduled payment preparation follows organizational timelines.
- Permissions restrict execution to authorized personnel only.

Topic 8: Memory & Context Management

Remembers:

- Invoice processing history and outcomes
- Vendor patterns and behaviors
- User approval patterns and preferences
- Exception resolution strategies
- Policy updates and changes

Forgets:

- Temporary processing contexts after completion
- Personal data after compliance period
- Low-importance operational details
- Outdated policy configurations

Structured Memory (DB Records)

- SQLite-based: Persistent storage for transactional data
- Normalized Schema: Efficient querying and indexing
- Audit Logs: Immutable records of all system actions

- Configuration Storage: Policy and rule persistence

Semantic Memory (Vector Store)

- Embedding-based: Contextual similarity matching
- Pattern Recognition: Learning from historical decisions
- Knowledge Graph: Entity relationships and connections
- Learning Persistence: Improving accuracy over time

Episodic Logs

- Session Tracking: Complete processing timelines
- Decision Trees: Rationale for all major decisions
- Exception Handling: Detailed context for edge cases
- Performance Metrics: Processing time and success rates

Learning Improvement Mechanism

- Feedback Loops: Incorporating outcome data into future decisions
- Pattern Recognition: Identifying successful processing strategies
- Adaptive Thresholds: Adjusting sensitivity based on results
- Continuous Training: Updating models with new examples

➤ Topic 9: Observability & Auditability

Structured Logs Per Agent Action

- Standardized Format: Consistent log structure across agents
- Correlation IDs: End-to-end traceability
- Performance Metrics: Processing time and resource usage
- Error Classification: Categorized failure types

Decision Traces

- Rationale Capture: Reasoning behind each decision
- Alternative Analysis: Explored alternatives and rejection reasons
- Confidence Scoring: Certainty levels for decisions
- Human Override Tracking: Manual interventions and reasoning

Evidence Storage

- Document Preservation: Original and processed document storage
- Metadata Capture: Complete context for each decision
- Chain of Custody: Complete handling history
- Backup & Recovery: Secure evidence preservation

Cost Tracking Per Workflow

- Resource Consumption: Compute, storage, and API costs
- Processing Efficiency: Cost per invoice processed
- ROI Analysis: Value delivered vs. operational costs
- Budget Controls: Automated cost management

➤ Topic 10: Data Models (High-Level)

Key Entities & Relationships

❖ Vendors

1. vendor_id (PK)
2. vendor_name
3. legal_name
4. tax_id
5. vat_number
6. bank_details (JSON)
7. contact_info (JSON)
8. payment_terms
9. status
(ACTIVE/BLOCKED/INACTIVE)
10. created_at, updated_at

❖ Invoices

1. invoice_id (PK)
2. vendor_id (FK)
3. invoice_number
4. invoice_date
5. due_date
6. amount
7. currency
8. po_number (optional, FK)
9. status(RECEIVED, EXTRACTED, VALIDATED, APPROVED, PAID, REJECTED)
10. created_at, updated_at

❖ Purchase Orders

1. po_id (PK)
2. po_number
3. vendor_id (FK)
4. po_date
5. total_amount
6. status(OPEN,CLOSED,CANCELLED)
7. line_items (JSON)
8. created_at, updated_at

❖ Receipts (Goods Receipts)

1. receipt_id (PK)
2. po_id (FK)
3. receipt_date
4. quantity_received
5. status (CONFIRMED, DISCREPANCIES)
6. discrepancies (JSON)
7. created_at, updated_at

❖ Approval States

1. approval_id (PK)
2. invoice_id (FK)
3. approver_user_id
4. approval_status (PENDING, APPROVED, REJECTED)
5. comments
6. created_at, updated_at

❖ Audit Logs

1. log_id (PK)
2. entity_type (INVOICE, VENDOR, APPROVAL, etc.)
3. entity_id
4. action_type (CREATE, UPDATE, DELETE, APPROVE, etc.)
5. user_id (if human action)
6. timestamp
7. details (JSON)

❖ Relationship Diagram

[Vendors] 1---* [Invoices]
[Vendors] 1---* [Purchase Orders]
[Purchase Orders] 1---* [Receipts]
[Invoices] 1---* [Approval States]
[Invoices] 1---* [Audit Logs]

➤ Topic 11: Failure Modes & Recovery

Realistic Failures

OCR Errors

- Detection: Confidence scores below threshold
- Recovery: Manual data entry with validation
- Prevention: Multiple OCR engine comparison

Missing Data

- Detection: Required fields not extracted
- Recovery: Supplier communication for missing information
- Prevention: Improved template recognition

Conflicting Policies

- Detection: Contradictory business rules identified
- Recovery: Human resolution and policy clarification
- Prevention: Policy conflict detection tools

Tool Downtime

- Detection: Service availability monitoring
- Recovery: Queue for retry when available
- Prevention: Redundant service providers

System Detection & Recovery

- Health Monitoring: Continuous system status checks
- Automated Recovery: Self-healing for common issues
- Escalation Procedures: Human intervention protocols
- Graceful Degradation: Continue operation with reduced functionality

➤ Topic 12: MVP Scope & What Is Deferred

Included in MVP

1. Invoice capture from email attachments
2. Basic OCR and data extraction
3. Simple 3-way matching (PO-Invoice)
4. Rule-based approval routing
5. Basic audit logging
6. Configuration management
7. Memory system with structured storage

Explicitly NOT Included

1. Advanced ML models for extraction accuracy
2. Real-time ERP integration (mock interfaces only)
3. Sophisticated semantic memory
4. Advanced fraud detection algorithms
5. Multi-language support
6. Mobile interface
7. Advanced reporting and analytics

Tradeoff Rationale

Focus on core autonomous processing capabilities first, then enhance with advanced features. This approach ensures a working system that can be iteratively improved based on real-world feedback.

